

Hybrid agreement: corpus results

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Loading data

Necessary libraries:

```
library(dplyr)

##
## Dołączanie pakietu: 'dplyr'
## Następujące obiekty zostały zakryte z 'package:stats':
##
##   filter, lag
## Następujące obiekty zostały zakryte z 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(ggplot2)
library(stringr)
library(lme4)
```

```
## Warning: pakiet 'lme4' został zbudowany w wersji R 4.4.2
```

```
## Ładowanie wymaganego pakietu: Matrix
```

Loading data:

```
numbers <- sprintf("%03d", 0:190)
filenames <- paste0("NKJP_300M_", numbers, ".csv")
filepaths <- file.path("C:/Users/magda/moje/hybrid-agreement-corpus/agreement-occurrences", filenames)
data <- lapply(filepaths, read.csv) %>% bind_rows()
rm(list=setdiff(ls(), "data"))
```

Preparing data

Mutating a boolean column for semantic agreement:

```
data %>%
  mutate(semantic_agr = case_when(source_other_gender == target_gender ~ 1,
                                   source_other_gender != target_gender ~ 0)) -> data
```

Giving the correct types of variables:

```
data %>%
  mutate(type_of_text = as.factor(type_of_text),
         source_cat = as.factor(source_cat),
```

```

source_gram_gender = as.factor(source_gram_gender),
source_other_gender = as.factor(source_other_gender),
target_gender = as.factor(target_gender),
type_of_agreement = as.factor(type_of_agreement)) -> data

```

Summary of the dataset:

```
summary(data %>% filter((source_cat != "profession") | (target_gender == "f")))
```

```

##      file_id      sentence_id      sentence
## Length:3559    Min.      :    1011    Length:3559
## Class :character 1st Qu.: 3680748    Class :character
## Mode  :character Median : 9098575    Mode  :character
##                      Mean  : 8943881
##                      3rd Qu.:12790382
##                      Max.   :19021548
##
##           type_of_text      source_cat    source_form
## typ_publ      :2212    depreciative:1040    Length:3559
## typ_lit       : 617    gendered      : 807    Class :character
## typ_fakt      : 226    profession   :1712    Mode  :character
## typ_qmow      : 143
## typ_net_interakt: 106
## typ_lit_proza  : 86
## (Other)       : 169
## source_lexeme    source_morph_descr source_gram_gender source_other_gender
## Length:3559      Length:3559        f : 100           f :2399
## Class :character Class :character m1:1714           m1:1160
## Mode  :character Mode  :character m2:1101
##                      m3: 274
##                      n : 370
##
##
## target_form      target_morph_descr target_gender    type_of_agreement
## Length:3559      Length:3559        f :1949          attributive:1873
## Class :character Class :character m1: 241          predicative:1478
## Mode  :character Mode  :character m2: 412          relpron      : 208
##                      m3: 462
##                      n : 495
##
##
##      distance      target_first      semantic_agr
## Min.      : 0.000    Min.      :0.0000    Min.      :0.0000
## 1st Qu.: 0.000    1st Qu.:0.0000    1st Qu.:0.0000
## Median : 1.000    Median :0.0000    Median :1.0000
## Mean   : 2.625    Mean   :0.4608    Mean   :0.5471
## 3rd Qu.: 2.000    3rd Qu.:1.0000    3rd Qu.:1.0000
## Max.   :802.000    Max.   :1.0000    Max.   :1.0000
##

```

Creating a separate table for each category of nouns. I got a lot of results with profession nouns and most of them didn't actually have a possibility of hybrid agreement, so I'm looking only at a subset of them. I will analyse them separately from the rest.

```
data %>%
  filter(source_cat == 'profession') %>%
  filter(target_gender == 'f') -> prof

data %>%
  filter(source_cat == 'gendered') -> gend

data %>%
  filter(source_cat == 'depreciative') -> depr
```

Descriptive statistics

Proportions of semantic agreement with gendered nouns and depreciative forms in each type of agreement that was considered: attributive, predicative, relative pronoun.

```
gend %>%
  group_by(type_of_agreement) %>%
  summarise(prop = sum(semantic_agr)/n()) -> gend_summary

depr %>%
  group_by(type_of_agreement) %>%
  summarise(prop = sum(semantic_agr)/n()) -> depr_summary

gend_summary
```

```
## # A tibble: 3 x 2
##   type_of_agreement prop
##   <fct>             <dbl>
## 1 attributive      0.0221
## 2 predicative      0.0667
## 3 relpron         0.0189
```

```
depr_summary
```

```
## # A tibble: 3 x 2
##   type_of_agreement prop
##   <fct>             <dbl>
## 1 attributive      0.107
## 2 predicative      0.268
## 3 relpron         0.196
```

Similar table for profession nouns. What percent of all instances of hybrid agreement is attributive, predicative and relative pronoun?

```
prof %>%
  count(type_of_agreement) %>%
  mutate(percent = 100 * n / sum(n)) -> prof_summary

prof_summary
```

```
##   type_of_agreement  n  percent
## 1      attributive 845 49.357477
## 2      predicative 758 44.275701
## 3        relpron 109  6.366822
```

For each agreement, is it more common to have target first or source first?

```
data %>%
  group_by(type_of_agreement) %>%
  summarise(prop = sum(target_first)/n())
```

```
## # A tibble: 3 x 2
##   type_of_agreement    prop
##   <fct>              <dbl>
## 1 attributive        0.735
## 2 predicative        0.302
## 3 relpron            0.000762
```

Mean distance for each agreement and each type of noun:

```
data %>%
  filter(source_cat != "profession") %>%
  group_by(type_of_agreement, source_cat) %>%
  summarise(avg = mean(distance)) -> dist_summary
```

```
## `summarise()` has grouped output by 'type_of_agreement'. You can override using
## the `groups` argument.
```

```
dist_summary
```

```
## # A tibble: 6 x 3
## # Groups:   type_of_agreement [3]
##   type_of_agreement source_cat    avg
##   <fct>              <fct>    <dbl>
## 1 attributive        depreciative 0.322
## 2 attributive        gendered    0.380
## 3 predicative        depreciative 1.62
## 4 predicative        gendered    1.94
## 5 relpron            depreciative 0.565
## 6 relpron            gendered    0.698
```

Based on the agreement source hierarchy, depreciatives should have less semantic agreement than gendered nouns. Is that so?

```
data %>%
  filter(source_cat != "profession") %>%
  group_by(source_cat) %>%
  summarise(prop = sum(semantic_agr)/n())
```

```
## # A tibble: 2 x 2
##   source_cat    prop
##   <fct>        <dbl>
## 1 depreciative 0.204
## 2 gendered     0.0285
```

Is there any noun among the profession nouns that doesn't have feminine agreement?

```
data %>%
  filter(source_cat == "profession") %>%
  group_by(source_lexeme) %>%
  summarise(prop = sum(semantic_agr)/n())
```

```
## # A tibble: 60 x 2
##   source_lexeme    prop
##   <chr>           <dbl>
```

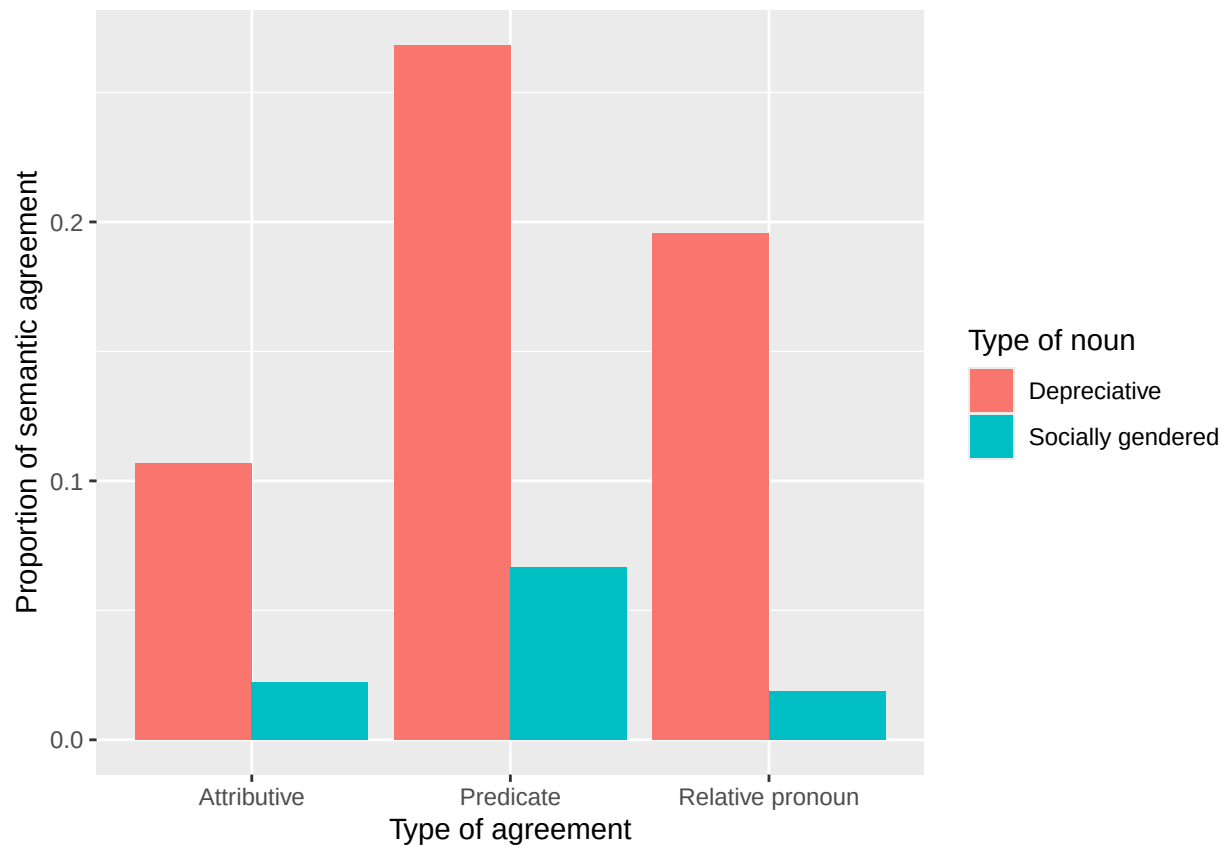
```
## 1 adiunkt      0.0341
## 2 architekt    0.00556
## 3 burmistrz   0.0167
## 4 chirurg      0.00181
## 5 demiurg      0
## 6 dramaturg    0
## 7 drań         0.00411
## 8 dupek        0.0110
## 9 dureń        0.0588
## 10 dziekan     0.032
## # i 50 more rows
```

Plots

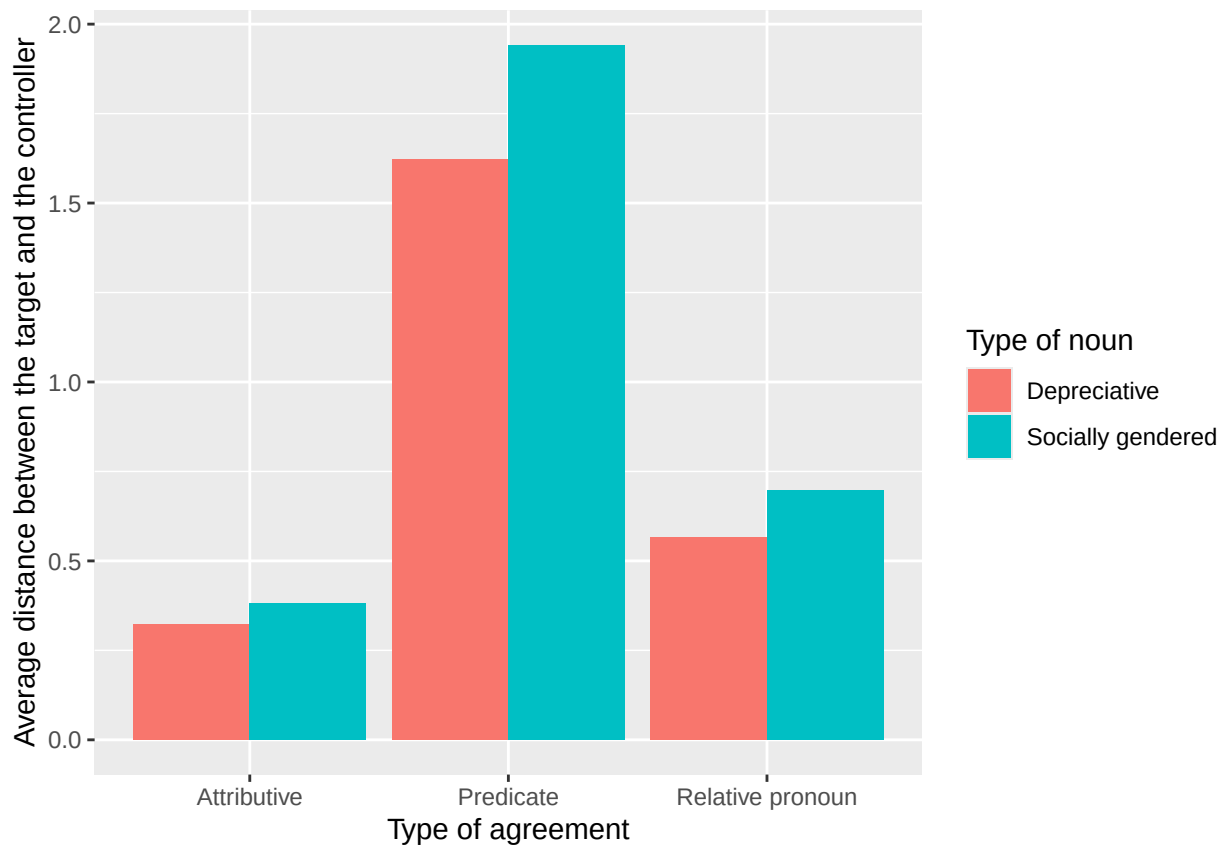
```
ggplot(data %>%
  filter(source_cat != "profession") %>%
  group_by(type_of_agreement, source_cat) %>%
  summarise(prop = sum(semantic_agr)/n())) +
  geom_col(aes(x = type_of_agreement,
    y = prop,
    fill = source_cat),
    position = "dodge") +
  labs(x = "Type of agreement",
    y = "Proportion of semantic agreement",
    fill = "Type of noun") +
  scale_x_discrete(labels = c("Attributive", "Predicate", "Relative pronoun")) +
  scale_fill_discrete(labels = c("Depreciative", "Socially gendered")) -> prop_plot
```

```
## `summarise()` has grouped output by 'type_of_agreement'. You can override using
## the `.groups` argument.
```

```
prop_plot
```



```
ggplot(dist_summary) +
  geom_col(aes(x = type_of_agreement,
               y = avg,
               fill = source_cat),
           position = "dodge") +
  labs(x = "Type of agreement",
       y = "Average distance between the target and the controller",
       fill = "Type of noun") +
  scale_x_discrete(labels = c("Attributive", "Predicate", "Relative pronoun")) +
  scale_fill_discrete(labels = c("Depreciative", "Socially gendered", "Profession")) -> dist_plot
dist_plot
```



Hypothesis testing

I realized that maybe I should exclude the profession nouns from this model, since there is a lot of examples that aren't actually hybrid agreement.

First I get a separate dataframe for models:

```
data -> data_for_model
data_for_model %>%
  filter(source_cat != "profession") -> data_for_model
```

I will start with a multivariate model, then do a few singular:

```
glm(semantic_agr ~ type_of_agreement + distance + target_first + source_cat,
    family = binomial(link = "logit"), data = data_for_model) -> multi_model
summary(multi_model)
```

```
##
## Call:
## glm(formula = semantic_agr ~ type_of_agreement + distance + target_first +
##      source_cat, family = binomial(link = "logit"), data = data_for_model)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -1.35979    0.21481  -6.330 2.45e-10 ***
## type_of_agreementpredicative  0.45490    0.21518   2.114  0.0345 *
## type_of_agreementrelpron    -0.21101    0.39788  -0.530  0.5959
## distance         0.03983    0.03242   1.229  0.2191
```

```
## target_first          -0.99116    0.20649  -4.800 1.59e-06 ***
## source_catgendered    -1.73335    0.23864  -7.263 3.77e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1407.8  on 1846  degrees of freedom
## Residual deviance: 1187.5  on 1841  degrees of freedom
## AIC: 1199.5
##
## Number of Fisher Scoring iterations: 6
```

Using a generalized linear model I want to see if type of agreement is a good predictor of whether the agreement will be semantic – the dependent variable is binary, so I'm using logistic regression. First I need to code the categorical variable so that the reference level is not chosen alphabetically. I'm using forward difference coding for the `type_of_agreement` variable, because I want to compare each level to the next one, to test the hypothesis that there's more semantic agreement in predicative agreement than in attributive and more in relative pronoun than in predicative.

```
forward.diff <- matrix(c(2/3, -1/3, -1/3, 1/3, 1/3, -2/3), ncol=2)
data_for_model$type_of_agreement <- as.factor(data_for_model$type_of_agreement)
contrasts(data_for_model$type_of_agreement) <- forward.diff
```

```
glm(semantic_agr ~ type_of_agreement,
    family = binomial(link = "logit"), data = data_for_model) -> model
```

```
levels(data_for_model$type_of_agreement)
```

```
## [1] "attributive" "predicative" "relpron"
```

```
summary(model)
```

```
##
## Call:
## glm(formula = semantic_agr ~ type_of_agreement, family = binomial(link = "logit"),
##      data = data_for_model)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -2.0740     0.1238 -16.758  <2e-16 ***
## type_of_agreement1 -1.6722     0.1631 -10.249  <2e-16 ***
## type_of_agreement2  1.0042     0.3449   2.911   0.0036 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1407.8  on 1846  degrees of freedom
## Residual deviance: 1284.3  on 1844  degrees of freedom
## AIC: 1290.3
##
## Number of Fisher Scoring iterations: 5
```

I also tried computing a model predicting only based on the distance. Here I'm not coding the variable, since it's not categorical.


```
glm(semantic_agr ~ distance,
    family = binomial(link = "logit"), data = data_for_model) -> dist_model
summary(dist_model)
```

```
##
## Call:
## glm(formula = semantic_agr ~ distance, family = binomial(link = "logit"),
##      data = data_for_model)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -2.07801    0.07966 -26.087  < 2e-16 ***
## distance     0.14115    0.02988   4.723 2.32e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1407.8  on 1846  degrees of freedom
## Residual deviance: 1386.5  on 1845  degrees of freedom
## AIC: 1390.5
##
## Number of Fisher Scoring iterations: 4
```

I also computed a model based on ordering of the target and source of agreement. The variable is equal to 1 when the target is before the source, 0 when it's not. Does this require any specific coding?

```
glm(semantic_agr ~ target_first,
    family = binomial(link = "logit"), data = data_for_model) -> order_model
summary(order_model)
```

```
##
## Call:
## glm(formula = semantic_agr ~ target_first, family = binomial(link = "logit"),
##      data = data_for_model)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -1.30833    0.08401 -15.574  <2e-16 ***
## target_first -1.53658    0.16216  -9.476  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1407.8  on 1846  degrees of freedom
## Residual deviance: 1301.9  on 1845  degrees of freedom
## AIC: 1305.9
##
## Number of Fisher Scoring iterations: 5
```

A model based on the type of noun, to see if split hybrids are less likely to have semantic agreement.

```
glm(semantic_agr ~ source_cat,
    family = binomial(link = "logit"), data = data_for_model) -> noun_type_model
summary(noun_type_model)
```

```
##
## Call:
## glm(formula = semantic_agr ~ source_cat, family = binomial(link = "logit"),
##      data = data_for_model)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -1.36243    0.07697 -17.700  <2e-16 ***
## source_catgendered -2.16649    0.22512  -9.624  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1407.8  on 1846  degrees of freedom
## Residual deviance: 1260.8  on 1845  degrees of freedom
## AIC: 1264.8
##
## Number of Fisher Scoring iterations: 6
```